TMNF · B01-RACE · CSCI218 FOUNDATIONS OF AI

CAN A SEARCH ALGORITHM BEAT MY TRACKMANIA PB?

An AI that drives the real game physics, tick by tick, until it finds a faster run

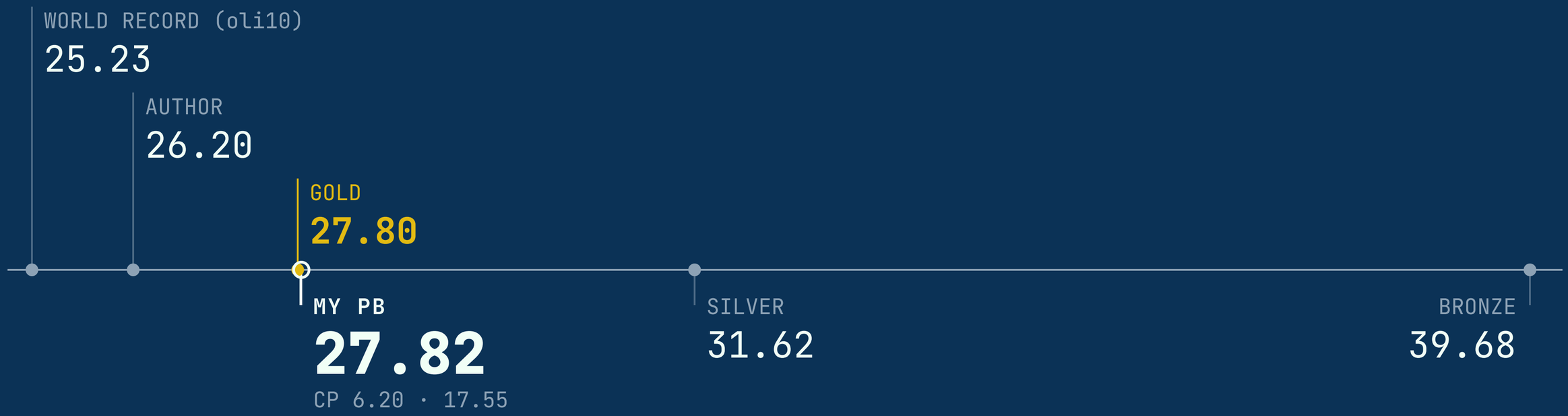
PB 27.82 → 27.28 · WR 25.23 → 25.21

[Team members] · UOWD · Autumn 2026

TRACKMANIA NATIONS FOREVER · B01-RACE

CAN AN AI BEAT MY PB?

0.02 S SHORT OF GOLD



Can an AI search, driving the real game physics, beat my PB and the gold medal?

FIND THE FASTEST SEQUENCE OF KEY PRESSES THROUGH ONE TRACK

PHYSICS TICK
10 ms

CONTROLS
4 keys

POSSIBLE RUNS
$\approx 10^{502}$

STATE
the full car state at one tick

ACTION
a key combination held for 50 ms

COST
race time

8 key combinations every 50 ms for 27.8 s: far too many to try, so the search needs a heuristic and pruning. The game is deterministic: the same state and inputs always give the same run, so a state can be saved, branched and replayed exactly.

THE REAL GAME IS THE SIMULATOR

01

GAME

TrackMania Nations Forever + TMInterface 2.2.1 via ModLoader. Physics in 10 ms ticks.

02

OUR PLUGIN

SearchBridge (AngelScript). Save and restore the car state, step the physics, stream frames, over TCP.

03

SEARCH

tmsearch (Python). Beam search and local search drive the plugin.

deterministic · bit-identical repeats · finish blocked, so no run ever counts as a record

27.820

MY PB, RE-DRIVEN TICK BY TICK

01 · PARSE

every key press, checkpoint and medal time from my replay file

02 · RE-DRIVE

the same inputs through the plugin give an identical 27.820, so our timing matches the game

03 · REFERENCE LINE

the car's position every 10 ms. Progress along it is the heuristic and is used for pruning

BEAM SEARCH THROUGH TIME

PART	DESIGN
Step	8 key combinations, each held 5 ticks (50 ms)
Beam	keep the best 40 states in each time layer
Score	progress along the reference line: ms ahead of my PB ghost, 0.3 s lookahead
Prune	off the line, 0.6 s behind, missed checkpoint, duplicates
Safety net	my own run rides along in every layer

RESULT AFTER 20 MIN

27.66

▼ -0.16 s vs PB · beats gold

177,163 expansions

A LEAD IN THE FIRST CORNERS, GONE A SECOND LATER

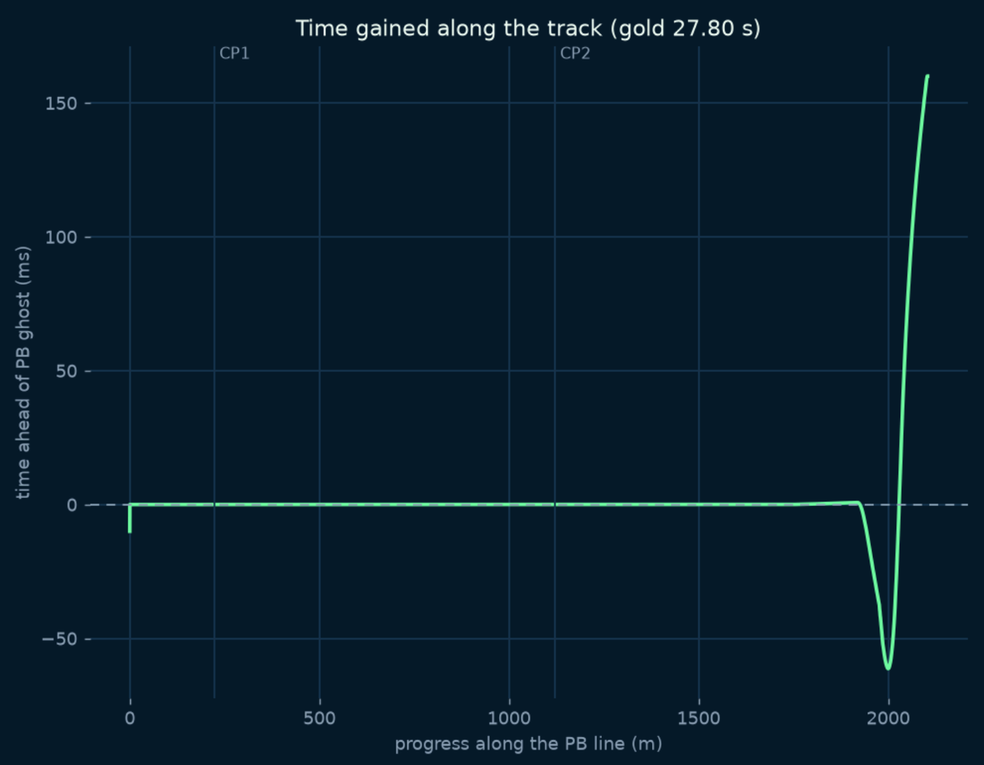
AT 5 S → AT 6 S

+45 ms → ±0 ms

AT 22 S → AT 23 S

+96 ms → ±0 ms

Myopic: it ranks by short-term progress. A 0.3 s lookahead rewards carrying speed into a corner, which costs time a moment later. Next: judge every change by the finish time.



Beam 27.66 vs my PB: the only gain is at the final drop

GOLD FELL IN 20 MINUTES. TOO EASY.



VIDEO · the AI's run with its live data panel

The gold ghost takes the early turns faster

→ time warp, and one worker per section of the track

It brakes before the last turns and slides

→ new moves: lift off the throttle instead of braking

The middle and the final jump look right

→ keep a change only if the whole run finishes faster

Human review of the AI's replays became new search moves, a new evaluation and more compute.

HILL CLIMBING ON THE KEY PRESSES

MOVE	WHAT IT DOES
Shift	move a key press or release by a few ticks
Tap	add a short tap
Lift	come off the throttle briefly
Brake → lift	coast instead of braking
Steer nudge / shift	adjust or move the analog steering

Kept only if the whole run finishes earlier. Rejected early once 150 ms behind.

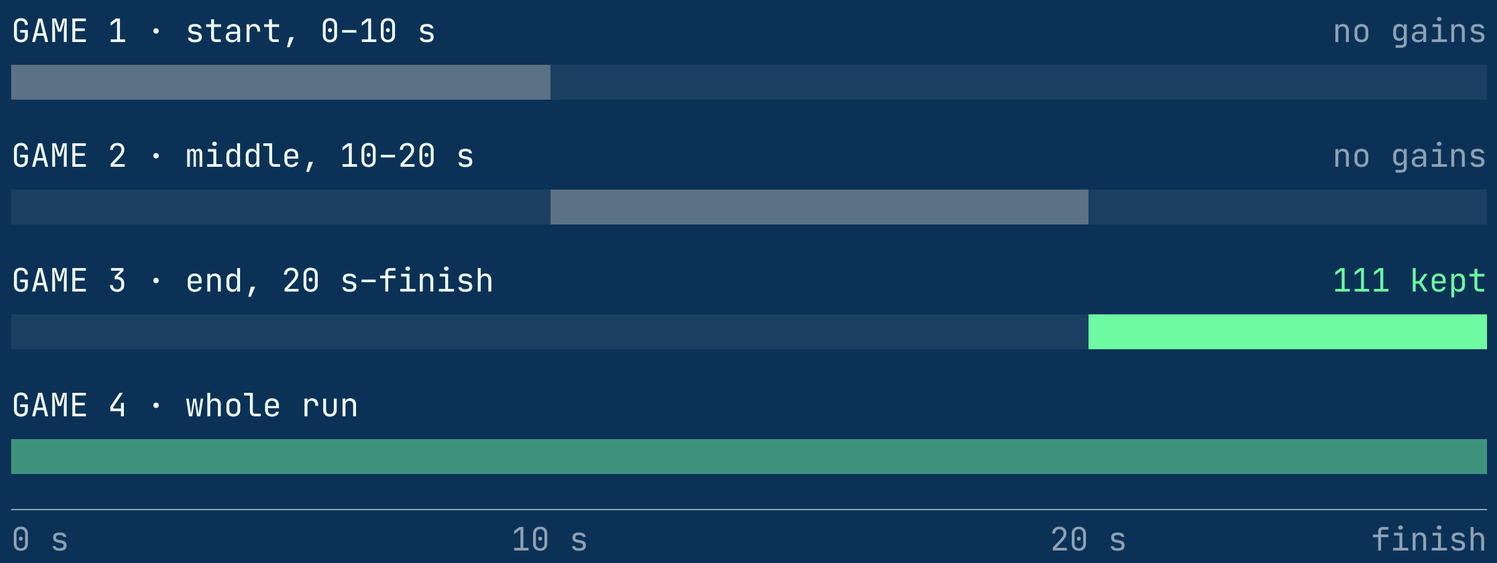
FROM 27.66, 30 MIN

27.42

▼ -0.24 s

4,723 evaluations · 125 accepted

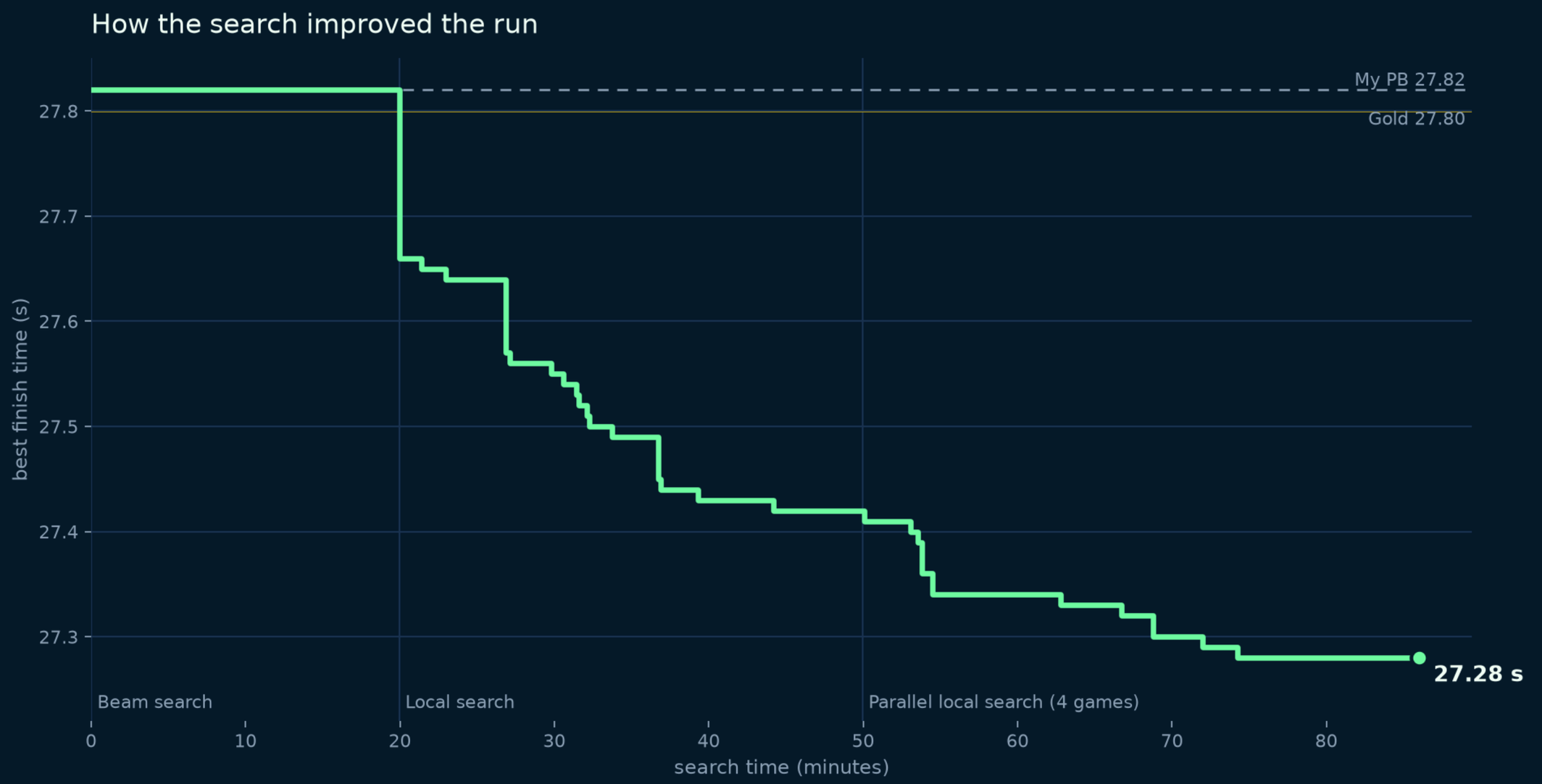
FOUR GAMES, ONE SHARED BEST RUN



27.28

▼ -0.54 s vs my PB

Any faster run is written to a shared best file and adopted by the others. Time warp re-aligns the rest of the run after an early change.



27.82 → 27.28 s in 86 minutes of search. Gold (27.80) fell in the first 20.

BACK ON THE THROTTLE SOONER, OFF IT LATER

SAVED	AT	CHANGE
30 ms	21.55 s	throttle back on 20 ms earlier
20 ms	25.50 s	stay on the throttle 40 ms longer
20 ms	25.62 s	stay on the throttle 40 ms longer
10 ms	23.85 s	release the brake 10 ms earlier
10 ms	25.60 s	a 20 ms throttle tap



VIDEO · the final drop in slow motion

The biggest kept tweaks. Each was found by trial in the game and kept only because the finish time dropped.

SEARCH ALGORITHMS COMPARED

Same game, same 8 actions held for 50 ms, same pruning · A* budget: 180,000 game steps (what beam width 40 needed to finish) · f = g + w·h, g = race time so far

DISTANCE
h = remaining reference-line length ÷ 1000 km/h (the game's speed cap). Admissible: it never overestimates.

PB TIME LEFT
h = the time my PB still needed from that point. Better informed, but not admissible.

METHOD	HEURISTIC	RESULT	GAME STEPS	TIME
Beam search, width 1 (greedy)	PB ghost ranking	27.82 (no gain)	11,182	1.5 min
Beam search, width 10	PB ghost ranking	27.82 (no gain)	49,534	6.5 min
Beam search, width 20	PB ghost ranking	27.82 (no gain)	92,174	12.2 min
Beam search, width 40	PB ghost ranking	**27.66**	177,163	19.9 min
A*	distance	did not finish: 3% of track (71 m)	180,000	23.5 min
Weighted A*, w = 5	distance	did not finish: 5% (100 m)	180,000	23.7 min
Weighted A*, w = 20	distance	did not finish: 11% (229 m)	180,000	23.4 min
A*	PB time left	did not finish: 9% (198 m)	180,000	24.2 min
Weighted A*, w = 1.5	PB time left	did not finish: 12% (258 m)	180,000	23.8 min
Weighted A*, w = 2	PB time left	did not finish: 11% (232 m)	180,000	25.4 min

Width 10 was run twice: identical inputs, time and step count, so the game and the search are deterministic.
The A* runs shared the CPU with another job, so game steps, not minutes, is the fair cost column.

A* never got past 12% of the track: the admissible h makes it explore breadth-first in time, weighting makes it dive into slow branches that pruning cuts, and the PB heuristic plateaus at f ≈ 27.82. Beam search bounds the work per 50 ms layer, but only width 40 beat my PB.

REPEATED RUNS: IS IT LUCK?

LOCAL SEARCH FROM MY PB · 5 SEEDS · 1,500 EVALUATIONS EACH (11–14 MIN EACH ON ONE GAME)

27.28 ± 0.14 s

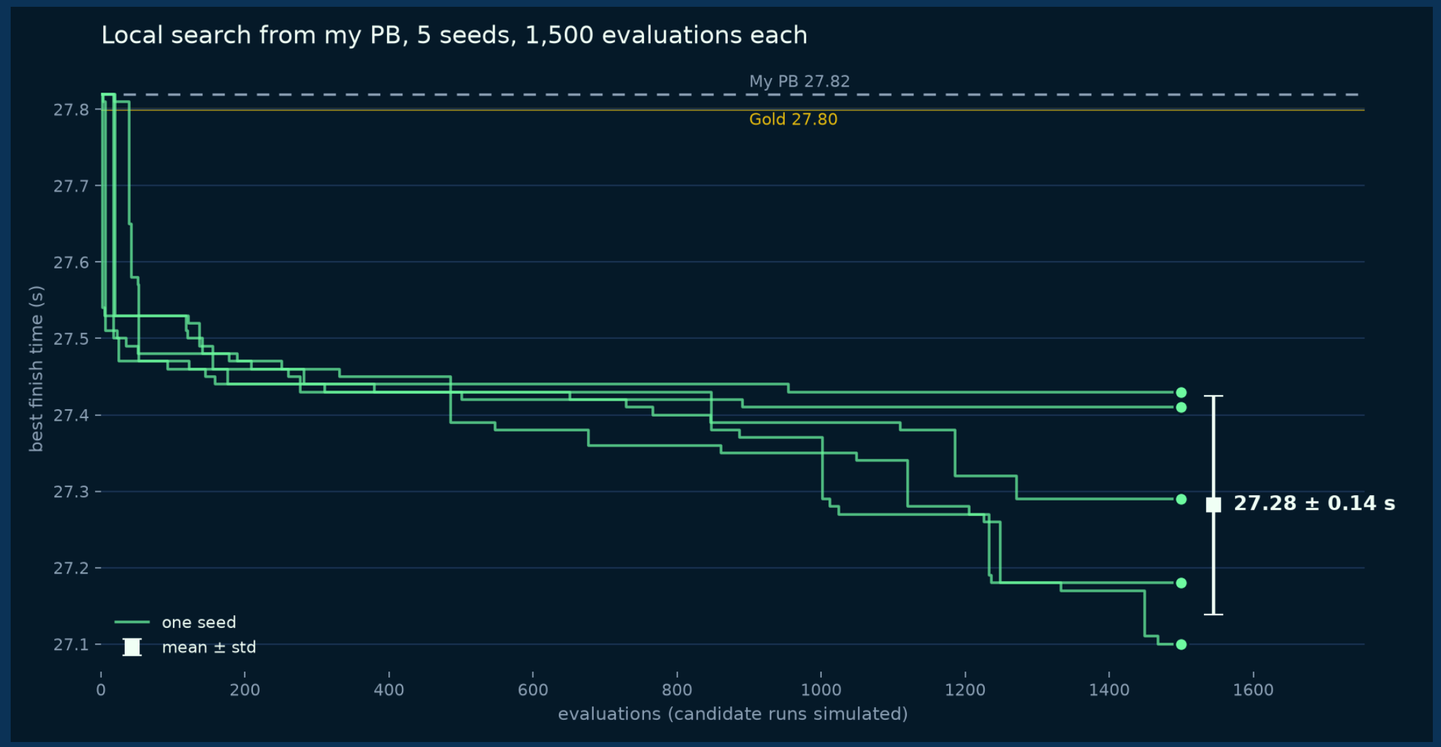
mean ± sample std

BEST
27.10

WORST
27.43

BEAT GOLD
5 of 5

AVERAGE GAIN
0.54 s
vs my PB



Reliable (every seed beat gold) but a wide spread: the best seed, 27.10 in 13 min on one game, beat the 86-minute four-game pipeline (27.28).

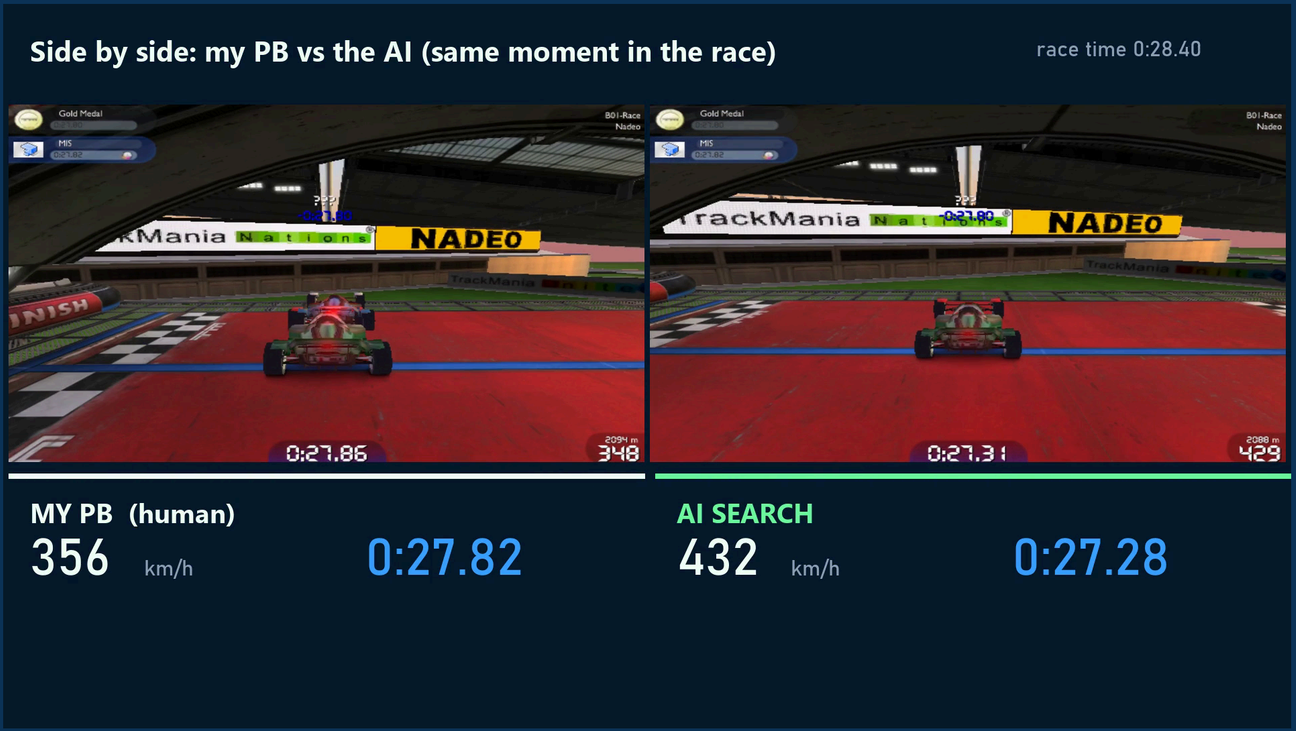
MAIN PIPELINE: THE RUN IN THE VIDEOS

27.28

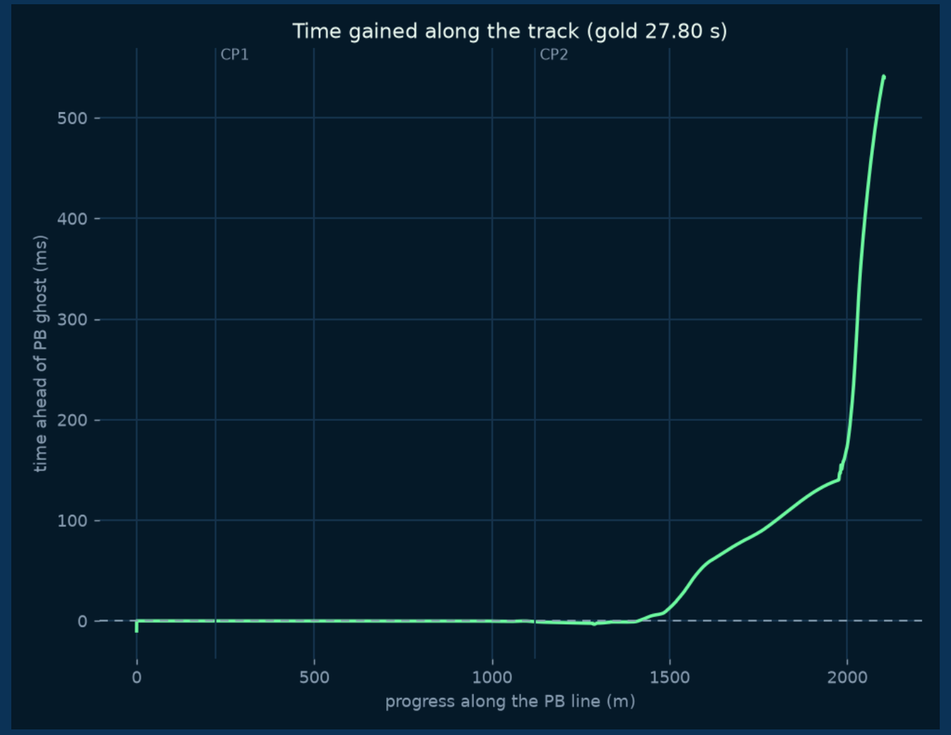
▼ -0.54 s vs my PB · beats gold

RUN	TIME
AI on the WR line	**25.21**
World record, oli10	25.23
Author medal	26.20
Local search from my PB, best of 5 seeds (13 min, 1 game)	**27.10**
AI from my PB: main pipeline (86 min, 4 games)	**27.28**
Local search	27.42
Beam search	27.66
Gold medal	**27.80**
My PB	27.82

SAME LINE, MORE SPEED AT THE END



VIDEO · my PB and the AI at the same moment in the race



Time gained on my PB ghost along the track: nothing early, then 0.54 s in the last corners and the drop

THE WORLD RECORD: MATCHED, THEN BEATEN

The last metres at 0.25x

race time 0:25.42



WORLD RECORD · oli10
365 km/h 0:25.23

AI ON HIS LINE
360 km/h 0:25.21

VIDEO · oli10's 25.23 vs the AI on his line

Tool-assisted and offline: not a record, never uploaded.

25.21

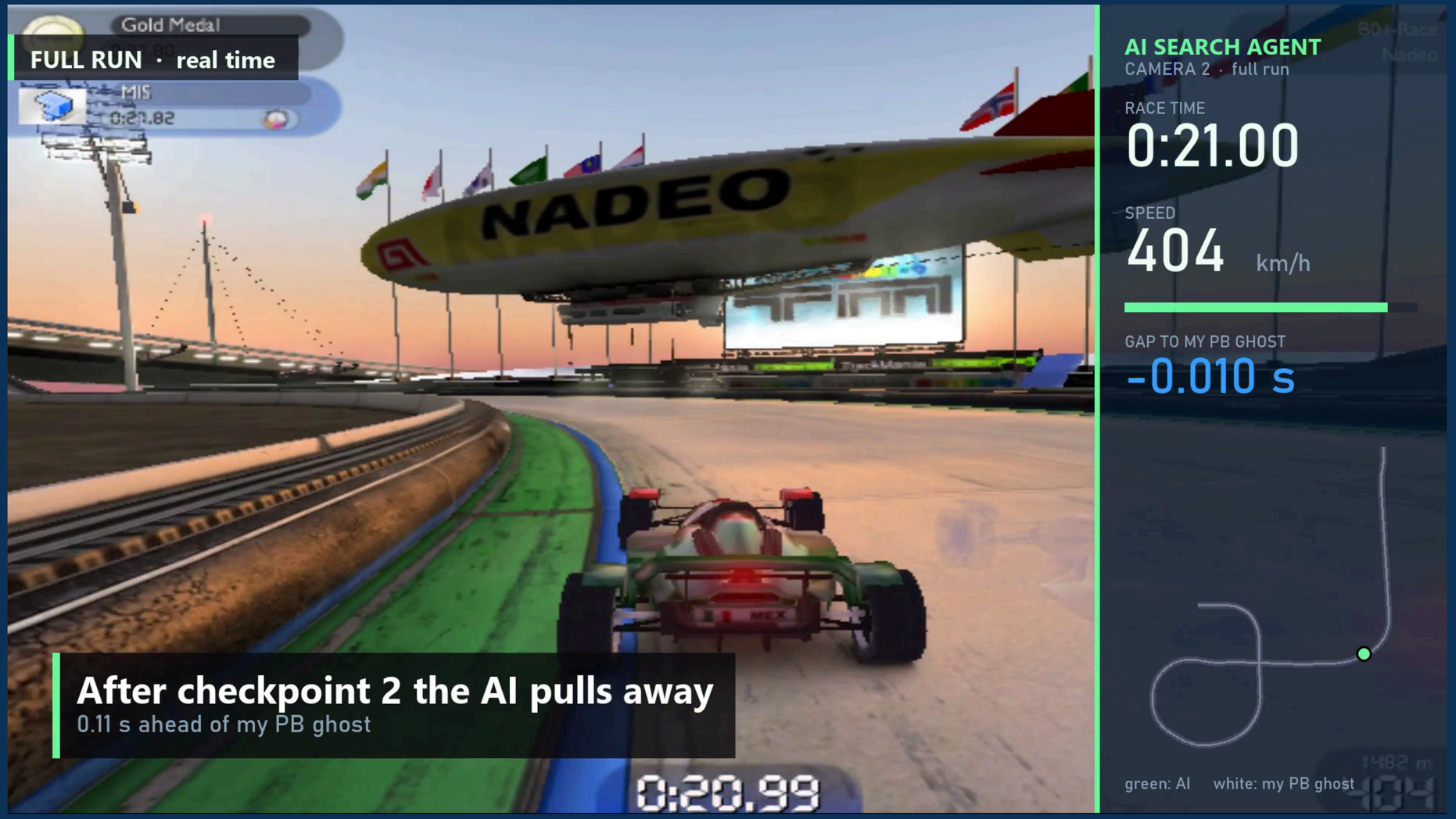
▼ -0.02 s vs 25.23, the human world record

MATCHED
his replay re-driven in our pipeline: 25.230, exact

BEATEN
after 941 tries (1.3 min): steer 3% more to the left for 280 ms at 19.69 s

FINAL
25.21 after 24,846 tries, 75 accepted, 35 min on 4 games

THE RUNS ON VIDEO



Recorded in the game, one frame per physics tick, from three cameras. Click to play (2 min).

WHAT THIS DOES NOT SHOW

ONE MAP

One track, one PB. No evidence yet that it generalises.

NO NEW ROUTES

Local search refines the line it starts from; from my line it is still 2.05 s off the WR.

TIED TO A LINE

Pruning around the reference line can cut out better lines.

PLUGIN-ONLY REPLAY

Runs replay exactly through our plugin, not as TMInterface scripts.

TOOL-ASSISTED

Offline only. Finishes are blocked, so nothing reaches a leaderboard.

27.82 → 27.28

Search inside the real game physics beat my best run by 0.54 s, then beat the human world record by 0.02 s starting from oli10's line.

MORE MAPS

test whether it generalises

NEW ROUTES

search that is not tied to one reference line

LEARN

train a policy from these runs

[Team members] · Questions?